

Chapter 7

Optimization: Choosing What Best Means

This chapter asks how mathematics can help compare possible choices. In optimization, a decision variable represents a choice, and an objective function assigns a numerical value to that choice. The objective might be something to maximize, such as profit or output, or something to minimize, such as travel time or energy use.

We use a business as our main example. The number of units produced and sold is the decision variable, and profit is the objective. Cost and revenue models let us build that objective; break-even and marginal analysis help interpret it. A good decision also depends on which choices are actually available and what the model assumes.

7.1 From Decisions to Objectives

An **objective function** measures an outcome for each possible choice. To optimize means to choose an available option that makes the objective as large or as small as the goal requires. The choice is represented by a **decision variable**.

Setting	Decision variable	Objective
Business	q , units produced and sold	Maximize profit $P(q)$
Trip planning	r , route selected	Minimize travel time $T(r)$
Energy use	s , system setting	Minimize energy $E(s)$

In each case, the goal must be stated before the choices can be compared. The objective assigns a value; it does not decide which goal matters most. This chapter develops profit as a familiar, concrete objective. The same way of thinking applies when the objective is cost, time, output, or another measurable benefit.

7.2 Cost Functions

The total cost of operating a business depends on two components: costs that are fixed regardless of production volume, and costs that vary with the number of units produced.

Definition. *Fixed cost.* **Fixed cost** F is the total cost incurred regardless of the quantity produced. Examples include rent, insurance, equipment, and administrative salaries.

Definition. *Variable cost.* **Variable cost** is the cost per unit of production, denoted v . The total variable cost at quantity q is $v \cdot q$.

Definition. *Total cost function.* The **total cost function** is:

$$C(q) = F + v \cdot q,$$

where F is fixed cost, v is variable cost per unit, and q is quantity produced.

The graph of $C(q)$ is a straight line with slope v and vertical intercept F . This reflects the fact that each additional unit adds the same amount, v , to the total cost.

Example 7.2.1. A candle maker pays \$180 per month in fixed costs and spends \$4 per candle in materials.

$$C(q) = 180 + 4q.$$

Candles q	Total cost $C(q)$
0	\$180
20	\$260
40	\$340
60	\$420
80	\$500
100	\$580

Each row adds \$80 to the cost, corresponding to 20 additional candles at \$4 each.

Example 7.2.2. (*Identifying cost components*) A food truck has the following weekly expenses: truck payment \$300, insurance \$50, ingredients \$3.50 per meal, packaging \$0.50 per meal.

Fixed cost: $F = 300 + 50 = \$350$.

Variable cost: $v = 3.50 + 0.50 = \$4.00$ per meal.

Total cost function: $C(q) = 350 + 4q$.

At $q = 200$ meals:

$$C(200) = 350 + 4(200) = 350 + 800 = \$1,150.$$

Exercises 7.2

7.2.1. Write the cost function $C(q)$ for each situation and compute the cost at the given quantity.

- (a) Fixed cost \$500, variable cost \$8 per unit, $q = 75$ units.
- (b) Fixed cost \$1,200, variable cost \$15 per unit, $q = 40$ units.
- (c) Monthly rent \$600, utilities \$100, ingredients \$2.50 per serving, $q = 300$ servings.

7.2.2. A bakery has fixed costs of \$250 per week and a variable cost of \$1.50 per loaf.

- (a) Write the cost function.
- (b) Fill in the cost table for $q = 0, 50, 100, 150, 200$.
- (c) By how much does the cost increase for each additional 50 loaves? What does this amount represent?

7.2.3. A company's total cost at $q = 100$ units is \$1,800 and at $q = 200$ units is \$2,600. Assume the cost function is linear.

- (a) Find the variable cost per unit.
- (b) Find the fixed cost.
- (c) Write the cost function $C(q)$.

7.2.4. A cost function is $C(q) = 400 + 6q$.

- (a) Identify the fixed cost and variable cost per unit.
- (b) Compute $C(0)$ and explain what it represents.
- (c) At what quantity does the total cost first exceed \$1,000?

7.3 Revenue Functions

Definition. *Revenue function.* The **revenue function** gives the total income from selling q units at price p per unit:

$$R(q) = p \cdot q.$$

Revenue is also an additive recursion. Each additional unit sold adds exactly p to revenue, starting from $R(0) = 0$. The graph of $R(q)$ is a straight line through the origin with slope p .

Example 7.3.1. The candle maker from Example 7.1.1 sells each candle for \$10.

$$R(q) = 10q.$$

Candles q	Revenue $R(q)$
0	\$0
20	\$200
40	\$400
60	\$600
80	\$800
100	\$1,000

Each row adds \$200 to revenue, corresponding to 20 additional candles at \$10 each.

Example 7.3.2. (*Finding the price from data*) A lemonade stand takes in \$0 when it sells nothing and \$120 when it sells 40 cups. What is the price per cup, and what is the revenue function?

$$p = \frac{R(q)}{q} = \frac{120}{40} = \$3.00 \text{ per cup.}$$

$$R(q) = 3q.$$

Exercises 7.3

7.3.1. Write the revenue function $R(q)$ for each situation and compute revenue at the given quantity.

- (a) Price \$25 per unit, $q = 60$ units.
- (b) Price \$4.50 per item, $q = 200$ items.
- (c) Price \$85 per service, $q = 12$ services.

7.3.2. A food truck earns \$560 selling 80 meals.

- (a) Find the price per meal.
- (b) Write the revenue function.
- (c) At what quantity does revenue reach \$1,400?

7.3.3. Two products are sold at different prices. Product A sells for \$12 per unit. Product B sells for \$20 per unit. In one week, 30 units of A and 15 units of B are sold.

- (a) Compute total revenue from each product.
- (b) Compute total revenue from both products combined.

7.3.4. A revenue function is $R(q) = 18q$. A cost function is $C(q) = 240 + 10q$. At $q = 40$:

- (a) Compute $R(40)$ and $C(40)$.
- (b) Is revenue greater or less than cost at this quantity?

7.4 Profit and Break-Even

Definition. *Profit function.* The **profit function** is revenue minus cost:

$$P(q) = R(q) - C(q) = pq - (F + vq) = (p - v)q - F.$$

When $P(q) > 0$, the business is profitable. When $P(q) < 0$, the business is operating at a loss. When $P(q) = 0$, the business breaks even.

Definition. *Break-even point.* The **break-even point** is the quantity q^* at which profit equals zero: $R(q^*) = C(q^*)$. Assume $p > v$, so each unit contributes a positive amount toward covering fixed costs. Under this condition,

$$pq^* = F + vq^* \Rightarrow (p - v)q^* = F \Rightarrow q^* = \frac{F}{p - v}.$$

If $F > 0$ and $p \leq v$, there is no nonnegative break-even quantity: each unit contributes no positive amount toward covering the fixed cost.

Example 7.4.1. The candle maker: $F = 180$, $v = 4$, $p = 10$.

$$P(q) = (10 - 4)q - 180 = 6q - 180.$$

q	$C(q)$	$R(q)$	$P(q)$
0	\$180	\$0	−\$180
20	\$260	\$200	−\$60
30	\$300	\$300	\$0
40	\$340	\$400	\$60
60	\$420	\$600	\$180

Break-even: $q^* = 180/6 = 30$ candles. At 30 candles the maker covers all costs exactly. At fewer than 30, a loss results. At more than 30, a profit results.

Suppose a venue allows the maker to choose only among $q = 20$, $q = 30$, and $q = 40$ candles. Their profits are −\$60, \$0, and \$60, respectively. If the goal is to maximize profit among these available choices, $q = 40$ is best. The break-even quantity marks where profit changes sign; it does not by itself identify the most profitable available choice.

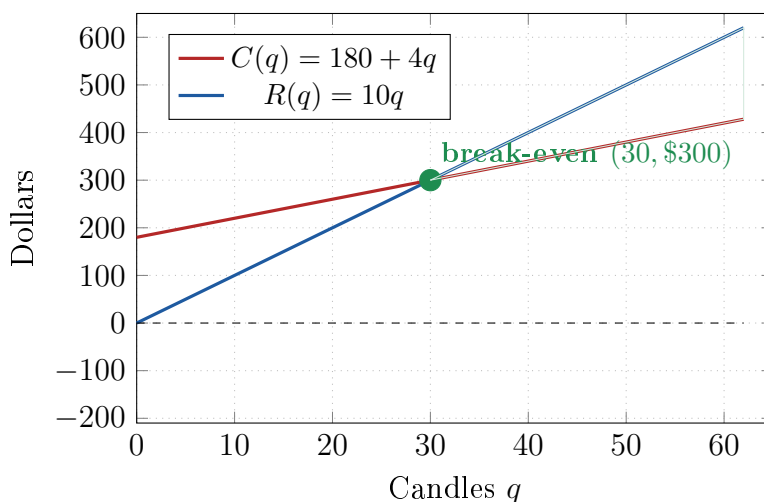
Example 7.4.2. A food truck: $F = 350$, $v = 4$, $p = 8$.

$$P(q) = (8 - 4)q - 350 = 4q - 350.$$

$$q^* = \frac{350}{8 - 4} = \frac{350}{4} = 87.5.$$

Since the truck cannot sell half a meal, the break-even quantity rounds up to $q^* = 88$ meals. At 88 meals:

$$P(88) = 4(88) - 350 = 352 - 350 = \$2 \text{ profit.}$$



Exercises 7.4

7.4.1. For each situation, write the profit function, find the break-even quantity, and compute the profit or loss at the given quantity.

- (a) $F = \$300$, $v = \$5$, $p = \$11$, find $P(80)$.
- (b) $F = \$600$, $v = \$8$, $p = \$14$, find $P(120)$.
- (c) $F = \$1,000$, $v = \$12$, $p = \$20$, find $P(100)$.

7.4.2. A student sells handmade greeting cards at a craft fair. The table fee is \$48 and each card costs \$1.50 to make. Cards sell for \$6 each.

- (a) Write $C(q)$, $R(q)$, and $P(q)$.
- (b) Find the break-even quantity.
- (c) Complete the profit table for $q = 0, 8, 10, 12, 16, 20$.
- (d) How much profit is earned at $q = 20$?

7.4.3. A company's profit function is $P(q) = 7q - 420$.

- (a) Find the break-even quantity.
- (b) Identify the fixed cost and the profit per unit beyond break-even.
- (c) At what quantity does profit reach \$280?

7.4.4. A business has fixed costs of \$500 and earns \$5 profit on each unit sold beyond break-even. At $q = 120$ units the business earns a profit of \$100.

- (a) Find the break-even quantity.
- (b) Find the price per unit and variable cost per unit, given that $p - v = \$5$.
Hint: use the formula for q^ .*

7.5 Marginal Analysis

Definition. *Marginal cost.* The **marginal cost** is the cost of producing one additional unit. For the linear cost function $C(q) = F + vq$, the marginal cost equals the variable cost per unit v .

Definition. *Marginal revenue.* The **marginal revenue** is the revenue from selling one additional unit. For the linear revenue function $R(q) = pq$, the marginal revenue

equals the price p .

Definition. *Marginal profit.* The **marginal profit** is the profit gained from selling one additional unit.

$$\text{Marginal profit} = p - v.$$

The marginal profit is the step size of the profit recursion. Each additional unit sold beyond break-even adds exactly $p - v$ to profit. This is the additive recursion from Chapter 1, now describing profit growth.

Example 7.5.1. For the candle maker: $p = 10$, $v = 4$.

$$\text{Marginal profit} = 10 - 4 = \$6 \text{ per candle.}$$

Every candle sold beyond the break-even point of 30 adds \$6 to profit.

At $q = 50$: the business is $50 - 30 = 20$ candles beyond break-even.

$$P(50) = 6 \times 20 = \$120.$$

Verification: $P(50) = 6(50) - 180 = 300 - 180 = \120 . ✓

Example 7.5.2. (*Marginal analysis at the boundary*) A print shop has $C(q) = 500 + 2q$ and $R(q) = 6q$.

$$\text{Marginal cost} = \$2, \quad \text{Marginal revenue} = \$6, \quad \text{Marginal profit} = \$4.$$

Break-even: $q^* = 500/4 = 125$ prints.

At $q = 124$: one unit below break-even.

$$P(124) = 4(124) - 500 = 496 - 500 = -\$4.$$

At $q = 125$: break-even.

$$P(125) = 4(125) - 500 = 0.$$

At $q = 126$: one unit above break-even.

$$P(126) = 4(126) - 500 = \$4.$$

Each unit past break-even adds exactly \$4 of profit, confirming that \$4 is the step size of the profit recursion.

Example 7.5.3. (*Comparing two pricing strategies*) A market vendor considers two pricing options for handmade soap.

	Strategy A	Strategy B
Price per bar	\$8	\$12
Variable cost	\$3	\$3
Fixed cost	\$200	\$200
Marginal profit	\$5	\$9
Break-even quantity	40	$22.2 \approx 23$

Strategy B reaches break-even faster. At $q = 40$:

$$P_A(40) = 5(40) - 200 = \$0, \quad P_B(40) = 9(40) - 200 = \$160.$$

Strategy B produces substantially more profit at the same volume. However, the higher price may reduce the number of buyers. Marginal analysis quantifies the profit per unit sold; it does not predict how many units will sell.

Exercises 7.5

7.5.1. Find the marginal cost, marginal revenue, and marginal profit for each situation.

(a) $C(q) = 300 + 7q$, $R(q) = 15q$

(b) $C(q) = 800 + 20q$, $R(q) = 35q$

(c) $F = \$450$, $v = \$9$, $p = \$14$

7.5.2. A craft fair vendor has marginal profit of \$6 per item and fixed costs of \$240.

(a) Find the break-even quantity.

(b) How much profit is earned at 10 items beyond break-even?

(c) At what quantity does profit reach \$120?

7.5.3. Two businesses have the same fixed cost of \$600. Business A has marginal profit \$10 per unit. Business B has marginal profit \$15 per unit.

(a) Find the break-even quantity for each.

(b) At $q = 80$ units, compute the profit for each.

(c) Which business benefits more from selling 10 additional units beyond $q = 80$? Compute the difference.

7.5.4. Explain in two complete sentences the connection between marginal profit and the additive recursion introduced in Chapter 1. Use a specific numerical example to support your explanation.

7.6 Graphical Analysis

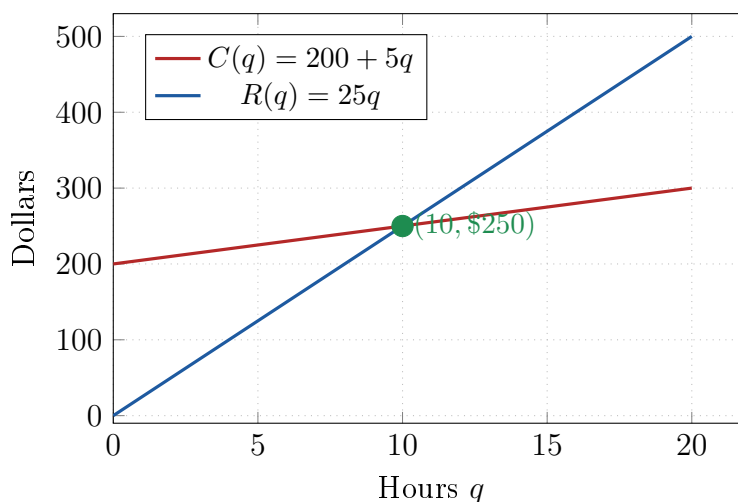
The cost and revenue functions are both straight lines. Their graph reveals the complete structure of the profit situation at a glance.

1. The **vertical intercept** of the cost line is the fixed cost F . The cost line never passes through the origin unless $F = 0$.
2. The **revenue line** always passes through the origin: at $q = 0$, revenue is zero.
3. The **break-even point** is the intersection of the two lines.
4. For quantities **to the left** of the break-even point, the cost line is above the revenue line: the business is at a loss.
5. For quantities **to the right** of the break-even point, the revenue line is above the cost line: the business is profitable.
6. The **slope** of the revenue line is the price p . The **slope** of the cost line is the variable cost v . The profit per additional unit is the difference in slopes: $p - v$.

Example 7.6.1. Graph the cost and revenue functions for a tutoring service: $F = \$200$, $v = \$5$ per hour, $p = \$25$ per hour.

$$C(q) = 200 + 5q, \quad R(q) = 25q.$$

Break-even: $q^* = 200/20 = 10$ hours.



The revenue line rises more steeply than the cost line because $p > v$. Beyond the intersection at $q = 10$ hours, revenue exceeds cost and the tutor earns a profit.

Example 7.6.2. (*Reading the graph*) A graph shows two lines. The cost line has vertical intercept \$400 and passes through the point (50, \$650). The revenue line passes through the origin and the point (50, \$750).

From the cost line:

$$v = \frac{650 - 400}{50 - 0} = \frac{250}{50} = \$5 \text{ per unit.}$$

$$C(q) = 400 + 5q.$$

From the revenue line:

$$p = \frac{750}{50} = \$15 \text{ per unit.}$$

$$R(q) = 15q.$$

Break-even:

$$q^* = \frac{400}{15 - 5} = \frac{400}{10} = 40 \text{ units.}$$

Exercises 7.6

7.6.1. For each pair of functions, identify the fixed cost, variable cost, price, and break-even quantity by reading the equations. Then sketch the graph, labeling both lines and the break-even point.

(a) $C(q) = 150 + 3q$, $R(q) = 9q$

(b) $C(q) = 400 + 10q$, $R(q) = 18q$

(c) $C(q) = 600 + 12q$, $R(q) = 20q$

7.6.2. A graph of cost and revenue functions shows the following:

- The cost line crosses the vertical axis at \$300.
- The cost line passes through (60, \$660).
- The revenue line passes through (60, \$900).

(a) Write $C(q)$ and $R(q)$.

(b) Find the break-even quantity.

(c) Find the profit at $q = 80$.

7.6.3. Two businesses sell the same product at the same price of \$20 per unit. Business A has fixed costs of \$800 and variable costs of \$8 per unit. Business B has fixed costs of \$400 and variable costs of \$12 per unit.

- (a) Write the profit function for each business.
- (b) Find the break-even quantity for each.
- (c) At $q = 60$, which business is more profitable? By how much?
- (d) Sketch both profit functions on the same axes. At what quantity do they produce equal profit?

7.7 The Observer and the Business

A cost, revenue, and profit analysis describes the same operation from the perspective of the numbers alone. Different observers read different quantities from that analysis because they are asking different questions.

Observer	Primary question
Business owner	How many units must I sell to cover costs?
Investor	At what volume does this become profitable?
Operations manager	What does each additional unit cost to produce?
Pricing strategist	How does a price change affect break-even?
Customer	What is the price per unit?

Each question is answered by a different quantity extracted from the same three functions. The break-even quantity answers the owner's question. The marginal profit answers the investor's question about per-unit profitability. The variable cost answers the operations manager's question. The price answers the customer's question.

A change in any one parameter propagates through all three functions. Raising the price p increases marginal profit, lowers the break-even quantity, and increases profit at every $q > q^*$. Raising the fixed cost F raises the break-even quantity without changing marginal profit. An observer focused on only one output may miss consequences visible to another.

Example 7.7.1. A food cart currently operates with: $F = \$350$, $v = \$4$, $p = \$9$.

$$q^* = \frac{350}{9 - 4} = 70 \text{ meals.}$$

At $q = 100$: $P(100) = 5(100) - 350 = \150 .

The owner considers two changes.

Option A: raise the price to \$11.

$$q_A^* = \frac{350}{11 - 4} = 50 \text{ meals. } P_A(100) = 7(100) - 350 = \$350.$$

Option B: reduce fixed cost to \$250 by moving to a smaller location.

$$q_B^* = \frac{250}{9 - 4} = 50 \text{ meals. } P_B(100) = 5(100) - 250 = \$250.$$

Both options lower the break-even point to 50 meals. At $q = 100$, Option A produces more profit (\$350 vs. \$250) because the marginal profit is higher. The investor asking about per-unit profitability would prefer Option A. The operations manager asking about overhead would prefer Option B. The customer asking about price would prefer neither.

Exercises 7.7

7.7.1. A business has $C(q) = 420 + 6q$ and $R(q) = 14q$.

- Find the break-even quantity and profit at $q = 80$.
- The owner raises the price to \$18. Find the new break-even quantity and profit at $q = 80$.
- The owner instead reduces fixed costs by \$120. Find the new break-even quantity and profit at $q = 80$ (using the original price of \$14).
- Write one sentence comparing the two options from the perspective of the break-even quantity.
- Write one sentence comparing the two options from the perspective of profit at $q = 80$.

7.7.2. A nonprofit sells event tickets to cover operating costs. Fixed costs are \$1,200. Variable costs are \$8 per attendee. Ticket price is \$20 per attendee.

- Find the break-even attendance.
- The nonprofit wants to reduce the ticket price to \$15 to attract more attendees. Find the new break-even attendance.
- Write one sentence explaining what the nonprofit must consider when choosing between the two ticket prices.

Chapter 7 Review

- A small printing company has monthly fixed costs of \$800 and a variable cost of \$3 per pamphlet. Pamphlets sell for \$8 each.

- (a) Write $C(q)$, $R(q)$, and $P(q)$.
 - (b) Find the break-even quantity.
 - (c) Complete the table for $q = 0, 80, 160, 200, 240$.
 - (d) Compute the profit at $q = 240$.
2. A mobile car wash has fixed costs of \$600 and charges \$25 per car. Supplies cost \$7 per car.
 - (a) Write all three functions and find the break-even quantity.
 - (b) Find the marginal profit and explain what it means in context.
 - (c) How many cars must be washed to earn a \$360 profit?
3. A cost function is $C(q) = 900 + 12q$ and a revenue function is $R(q) = 21q$.
 - (a) Find the break-even quantity.
 - (b) Sketch the graph, labeling the vertical intercept, the break-even point, and the slopes of both lines.
 - (c) What is the profit at $q = 150$?
4. A freelance designer charges \$80 per hour and has fixed monthly costs of \$320. The designer's time costs \$20 per hour (opportunity cost and materials).
 - (a) Write the profit function.
 - (b) Find the break-even number of hours.
 - (c) How many hours must be billed to earn \$1,000 profit per month?
5. A business currently sells a product for \$16 with variable cost \$7 and fixed cost \$540. A competitor enters the market, forcing the price down to \$13.
 - (a) Find the original break-even quantity and profit at $q = 80$.
 - (b) Find the new break-even quantity and profit at $q = 80$ with the lower price.
 - (c) By how much does the break-even quantity increase as a result of the price decrease?
6. Two bakeries both sell bread for \$7 per loaf. Bakery A: $F = \$400$, $v = \$2$. Bakery B: $F = \$200$, $v = \$4$.
 - (a) Write the profit function for each.
 - (b) Find the break-even quantity for each.
 - (c) At what quantity do the two bakeries earn equal profit? (Set $P_A(q) = P_B(q)$ and solve.)
 - (d) Which bakery is more profitable at high volume? Explain using marginal profit.
7. Explain in three complete sentences how the cost, revenue, and profit functions in this chapter are each examples of the additive recursion from Chapter 1. Identify the starting value and step size for each.